

B.Sc. Semester-II Examination, 2022-23**MATHEMATICS [Honours]**

Course ID : 22114

Course Code : SH/MTH/203/GE-2

Course Title : Algebra

[NEW SYLLABUS]

Time : 2 Hours

Full Marks : 40

*The figures in the right-hand margin indicate marks.**Candidates are required to give their answers in their own words as far as practicable.**Notations and symbols have their usual meaning.***UNIT-I**1. Answer any **five** from the following questions:

$$2 \times 5 = 10$$

- a) Find the number of positive and negative roots of the equation $x^5 - x^4 + x^3 + 8x^2 + 2x - 2 = 0$.
- b) Prove that if $x^2 = e$ for all element of a group G , then G is a commutative.
- c) Z is a Variable complex number such that an amplitude of $\frac{Z-i}{Z+1}$ is $\frac{\pi}{4}$. Show that the point Z lies on a circle in the complex plane.

[Turn Over]

- d) Show that for any three real numbers a, b, c , $a^3 + b^3 + c^3 \geq a^2b^2c^2(ab + bc + ca)$.
- e) Let W be the subspace of $\mathbb{R}^3$ defined by $W = \{(x, y, z) \in \mathbb{R}^3 : 2x + 2y + z = 0, 3x + 3y - z = 0, x + y - 3z = 0\}$. Find the dimension of W .
- f) If a is prime to b , prove that a^2 is prime to b and b^2 .
- g) If α, β and γ be the roots of the equation $x^3 + 2x^2 - 3x - 1 = 0$. Find the value of $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha}$.
- h) The eigen values of the matrix A are 5 and 3. Find the eigen values of the matrix $A^2 - 8A + 15I + 30A^{-1}$.

UNIT-II

2. Answer any four from the following questions:

$$5 \times 4 = 20$$

- a) i) Let $\rho = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid 3a + 4b = 7n, \text{ for some } n \in \mathbb{Z}\}$. Show that ρ is an equivalence relation.
- ii) Use Principle of mathematical induction to prove that $2 \cdot 7^n + 3 \cdot 5^n - 5$ is divisible by 24 for all $n \in \mathbb{N}$.

$$3 + 2 = 5$$

- b) i) Let $a_i > -\frac{1}{3}$, ($i = 1, 2, 3$) and $a + b + c = 1$.
Apply Cauchy-Schwarz inequality to prove
that $\sqrt{3a+1} + \sqrt{3b+1} + \sqrt{3c+1} \leq 3\sqrt{2}$.
- ii) Find the minimum value of $3x + 2y$ where
 x, y are positive real numbers satisfying the
condition $x^2y^3 = 48$. 3+2=5
- c) If α, β, γ be the roots of the equation
 $x^3 - px^2 + r = 0$, find the equation whose roots
are $\frac{1}{\alpha^2} + \frac{1}{\beta^2} - \frac{1}{\gamma^2}, \frac{1}{\beta^2} + \frac{1}{\gamma^2} - \frac{1}{\alpha^2}, \frac{1}{\gamma^2} + \frac{1}{\alpha^2} - \frac{1}{\beta^2}$.
- d) If H and K are subgroups of the group $(G, 0)$
then HK is subgroup of G , iff $HK = KH$.
- e) Reduce the reciprocal equation
 $x^5 - 6x^4 + 7x^3 + 7x^2 - 6x + 1 = 0$ to its standard
form and solve it.
- f) Find the linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ which maps
the basis vectors $(0, 1, 1), (1, 0, 1), (1, 1, 0)$ of
 $\mathbb{R}^3$ to the vectors $(2, 0, 0), (0, 2, 0), (0, 0, 2)$ of
 $\mathbb{R}^3$ respectively. Find $\text{Ker } T$ and $\text{Im } T$. Show that
 $\dim \text{Ker } T + \dim \text{Im } T = \dim \mathbb{R}^3$. 3+2=5

UNIT-III

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3. Answer any one of the following questions:

10×1=10

B

a) i) Let $f: \mathbb{R} \rightarrow S = \{x \in \mathbb{R} : 1 < x < 1\}$ defined by $f(x) = \frac{x}{1+|x|}$, $x \in \mathbb{R}$. Find f^{-1} .

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ii) Apply Descartes rule of signs to ascertain the minimum number of complex roots of the equation $x^6 - 3x^2 - 2x - 3 = 0$.

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iii) Find a matrix P and a diagonal matrix D such

that $P^{-1}AP = D$, where $A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$.

3+2+5=10

b) i) If α, β, γ be the roots of the equation $x^3 + px^2 + qx + r = 0$, find the value of $\sum \alpha^3 \beta^3$.

ii) Show that $\text{Log}(1+i)^3 \neq 3\text{Log}(1+i)$.

iii) Show that the ratio of the principal values of $(1+i)^{1-i}$ and $(1-i)^{1+i}$ is

$\sin(\log 2) + i \cos(\log 2)$. 4+2+4=10

B.Sc. 2nd Semester (G.E.), Internal Exam, 2022-23

Subject: Mathematics Course Title: Algebra.

F.M = 10 marks.

Time: 30 min.

Answer any two questions:

$$\underline{5 \times 2 = 10}$$

(1) If z be a complex number and $\frac{z+1}{z-i}$ be purely imaginary, then show that z lies on the circle whose centre is at $\frac{1}{2}(-1+i)$ and the radius is $\frac{1}{\sqrt{2}}$.

(2) State the principle of Mathematical Induction and using it to show that $(5^n - 1)$ is divisible by 4 for all positive integers n .

(3) Find the Echelon form of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 4 & 6 & 8 \end{bmatrix}$

and hence find the rank of A .

(4) If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be such that $T(a, b) = (a, 0)$ then show that T is a linear transformation.